• EQUAZIONI ALLE D.P.

PDE LINEARI 2° ORDINE

$\Delta = B^2 - 4AC$ $\qquad$ $Au_{xx} + Bu_{xy} + Cu_{yy} + \dots = 0$

CANONICHE

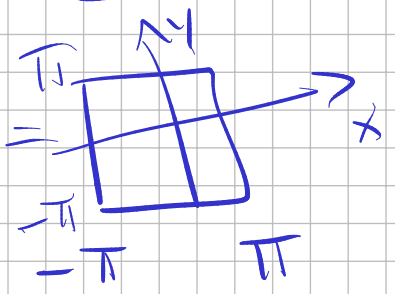
$\Delta > 0$ IPERBOLICHE $\qquad$ $\partial_{xy} u = 0$

$\Delta = 0$ PARABOLICHE $\qquad$ $\partial_{xx} u = 0$

$\Delta < 0$ ELLITTICHE $\qquad$ $A(\partial_{xx} u + \partial_{yy} u) = 0$

• SOLUZIONE EQUAZ. PDE ELLITTICHE

NEL CASO PERIODICO

$\Omega =$ 

$\Omega = [-\pi, \pi]^2$

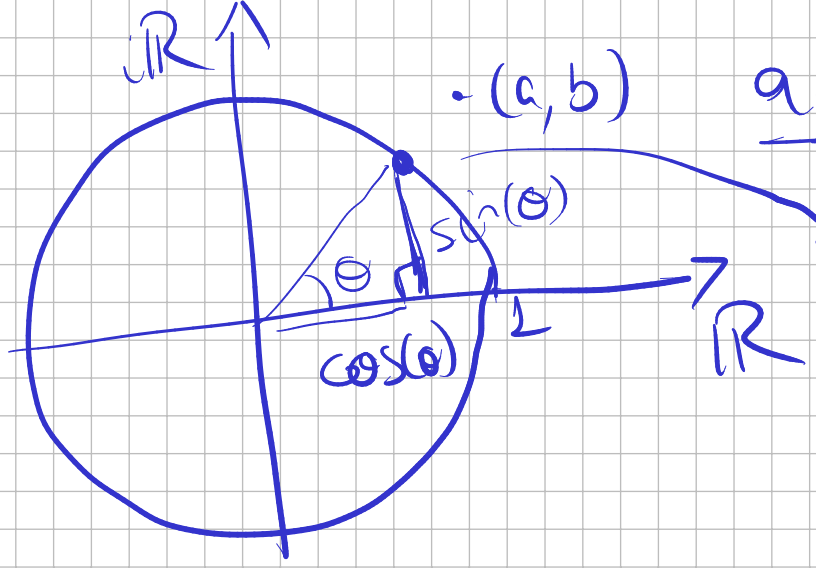
$u(-\pi, y) = u(\pi, y) \quad \forall\, y \in [-\pi, \pi]$

$u(x, \pi) = u(x, -\pi) \quad \forall\, x \in [-\pi, \pi]$

→ FOURIER

$\boxed{e^{ix}} = \cos(x) + i \sin(x)$

$\sin(x) = \dfrac{e^{ix} - e^{-ix}}{2i}$ $\qquad$ $\cos(x) = \dfrac{e^{ix} + e^{-ix}}{2}$



$a+ib$

$\cos(\theta)+i\sin(\theta)$

$x = a+ib$

$a\in\mathbb{R},\ b\in\mathbb{R}$

$$e^{ix} = e^{i(a+ib)} = e^{ai-b} = \underbrace{e^{-b}}\ \underbrace{e^{ai}}$$

$$\boxed{e^{a+ib} = \underbrace{e^{a}}_{\text{RAGGIO}}\ \underbrace{e^{ib}}_{b\ \text{ANGOLO}}}$$

$$\to\ \partial_x \underbrace{e^{ixk}}_{\text{AUTOVETT.}} = \underbrace{\boxed{ik}}_{\text{AUTOVALORE}}\ \underbrace{e^{ikx}}_{\to\ \text{AUTOVETT.}}$$

$k\in\mathbb{Z}$

$$\partial_{xx} e^{ikx} = (ik)^2 e^{ikx} = -k^2 e^{ikx}$$

$$\partial_x \sin(kx) = k\cos(kx)$$

$$\to\ \partial_{xx} \underbrace{\sin(kx)} = -k^2 \underbrace{\sin(kx)}$$

- $g(x)$ PERIODICA

$$g(x) = \sum_{k\in\mathbb{Z}} \hat{g}_k e^{ikx}$$

$$\hat{g}_k = \frac{1}{2\pi}\int_{-\pi}^{\pi} g(x)\cdot e^{-ikx}\,dx$$

<u>PARSEVAL</u> $\quad \|\hat{g}\|_2^2 = \sum_{k\in\mathbb{Z}} |\hat{g}_k|^2$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} |g(x)|^2\,dx = \frac{1}{2\pi}\|g\|_2^2$$

- PDE $\quad -u_{xx} - u_{yy} = g \qquad g$ PERIODICA CONTINUA

$u : \Omega \to \mathbb{R} \qquad g : \Omega \to \mathbb{R}$, $\Omega = [-\pi,\pi]^2$

1D $\quad u = \sum_k e^{ikx}\,\hat{u}_k$

2D $\quad u(x,y) = \sum_{k\in\mathbb{Z}}\sum_{j\in\mathbb{Z}} e^{ikx}e^{ijy}\,\underbrace{\hat{u}_{kj}}_{\text{VOGLIAMO TROVARLO COME!}}$

$$g(x,y) = \sum_{k,j\in\mathbb{Z}} e^{ikx}e^{ijy}\,\hat{g}_{kj}$$

$\hat{g}_{kj} \in \mathbb{C}$

(1) $-u_{xx} - u_{yy} = g$

$$u_{xx} = \sum_{k,j\in\mathbb{Z}} -k^2 e^{ikx}e^{ijy}\,\hat{u}_{kj}$$

$$u_{yy} = \sum_{k,j\in\mathbb{Z}} -j^2 e^{ikx}e^{ijy}\,\hat{u}_{kj}$$

$$(1) \quad -\sum_{k,j\in\mathbb{Z}} \hat{u}_{kj}(-k^2-j^2)\, e^{ikx} e^{ijy} \overset{!}{=} \sum_{k,j\in\mathbb{Z}} \hat{g}_{kj}\, e^{ikx} e^{ijy}$$

$\{e^{ikx}\}_{k\in\mathbb{Z}}$ sono ortogonali

$$\Rightarrow \quad \hat{u}_{kj}(k^2+j^2) = \hat{g}_{kj}$$

$$\boxed{\hat{u}_{kj} = \frac{\hat{g}_{kj}}{k^2+j^2}} \qquad k^2+j^2 \neq 0$$

$\hat{u}_{00} \rightarrow e^0 e^0 = 1$

<ISONO

$$\langle a,b\rangle := \int_{-\pi}^{\pi} a(x)\,\overline{b}(x)\,dx \qquad \overline{c+id} := c-id \quad c,d\in\mathbb{R}$$

$$\langle e^{ikx}, e^{ijx}\rangle = \begin{cases} 2\pi & k=j \\ 0 & k\neq j \end{cases}$$

$$\int_{-\pi}^{\pi} e^{ikx} e^{-ijx}\,dx = \int_{-\pi}^{\pi} e^{i(k-j)x}\,dx$$

$$= \begin{cases} k=j & = \int_{-\pi}^{\pi} e^{i\cdot 0\cdot x} = \int_{-\pi}^{\pi} 1\,dx = 2\pi \\ k\neq j & \int_{-\pi}^{\pi} e^{i(k-j)x} = \left[\frac{e^{i(k-j)x}}{i(k-j)}\right]_{-\pi}^{\pi} \end{cases}$$

$k \neq j$

$$= 0 \qquad \frac{1}{i(k-j)} \left[\cos\left(\underbrace{(k-j)}_{\in \mathbb{Z}} x\right) + i \sin((k-j)x) \right]_{-\pi}^{\pi}$$

periodica !

$= 0$

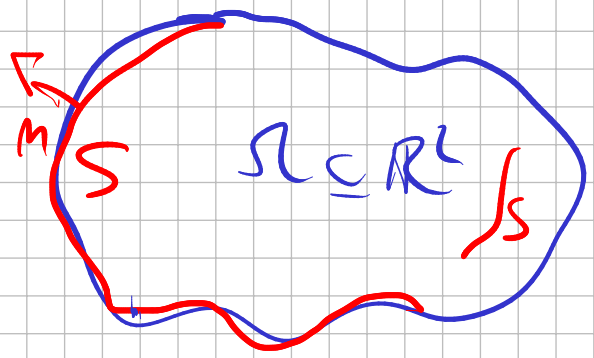
DEF PROBLEMA DI CAUCHY

PDE ORDINE k DOMINIO $\Omega \subset \mathbb{R}^d$

e S superficie di dim $d-1$

n NORMALE A S

$\boxed{\forall x \in S}$



$$\left.\begin{aligned} u(x) &= f_0(x) \\ \partial_n u(x) &= f_1(x) \\ &\vdots \\ \partial_n^{k-1} u(x) &= f_{k-1}(x) \end{aligned}\right\}$$

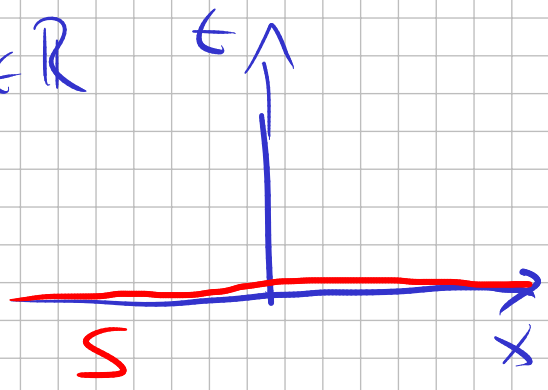
CONDIZIONI INIZIALI (CONDIZ. AL BORDO)

$$\partial_n u(x) := \lim_{\varepsilon \to 0} \frac{u(x+\varepsilon n) - u(x)}{\varepsilon}$$

ES. EQ. TRASPORTO + IC.

CONDIZ. INIZIALI

$$\text{I.C.}\begin{cases} u_t + u_x = 0 & (t,x) \in \Omega = \mathbb{R}^+ \times \mathbb{R} \\ u(0,x) = \sin(x) & \forall x \in \mathbb{R} \end{cases}$$



$S = \{(t,x) : t = 0\} \subseteq \mathbb{R}^2$

dimensione di $S = 1$

(È UNA RETTA!)

$u(t,x) = f(x-t)$ $\qquad f : \mathbb{R} \to \mathbb{R}$

FUNZIONE ARBITRARIA DA DETERMINARE

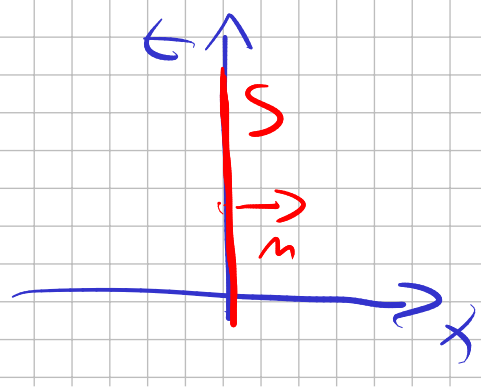
$u(0,x) = \sin(x) = f(x-0) = f(x)$

$u(t,x) = \sin(x-t)$

<u>ES2</u> PROBLEMA DI CAUCHY EQ. ONDE con BC

BOUNDARY CONDITIONS

$$\begin{cases} \partial_{tt} u - \partial_{xx} u = 0 & (t,x) \in \Omega = \mathbb{R}^+ \times \mathbb{R} \\ u(t,0) = \sin(t) & t \in \mathbb{R}^+ \\ u_x(t,0) = 0 & \forall t \in \mathbb{R}^+ \end{cases}$$



$S = \{(t,x) : x = 0\} \subset \Omega$

$\partial_{tt} u = \partial_x u$

ARBITRARIE
$f, g : \mathbb{R} \to \mathbb{R}$

$$u(t,x) = f(x-t) + g(x+t)$$

SOLUZ. GENERICA PDE

$$\begin{cases} u(t,0) = \sin(t) = f(-t) + g(t) & (2) \\ u_x(t,0) = 0 = f'(-t) + g'(t) & (3) \end{cases}$$

$$z(t) = f(-t)$$

$$z'(t) = -f'(-t)$$

(3) $-z'(t) + g'(t) = 0 \Rightarrow z'(t) = g'(t)$

$$z(t) = g(t) + C \qquad C \in \mathbb{R}$$

$$f(t) = g(-t) + C \qquad C \in \mathbb{R}$$

$$z(t) + g(t) = \sin(t)$$

$$2g(t) + C = \sin(t)$$

$$g(t) = \frac{\sin(t) - C}{2}$$

$$z(t) = \frac{\sin(t) + C}{2} \quad \left(= g(t) + C \right)$$

$$f(t) = \frac{\sin(-t) + C}{2}$$

$$u(t,x) = f(x-t) + g(x+t)$$

$$= \frac{\sin(t-x) + \cancel{C}}{2} + \frac{\sin(x+t) - \cancel{C}}{2}$$

$$= \frac{1}{2}\left[\sin(t-x) + \sin(x+t)\right]$$

1 SOLUZIONE ☺

TEOREMA CAUCHY-KOVALEVSKAYA

PROBLEMA DI CAUCHY

PER PDE <u>LINEARE</u>

- $x^0 \in S$ $\qquad$ S È ANALITICA
- S IN x^0 NON È SUPERF. CARATTERISTICA
- i COEFF della PDE, g, $f_0, \dots f_{k-1}$ ANALITICI

$\Rightarrow \exists!\, u$ sol. AL PROBLEMA DI CAUCHY ed è ANALITICA.

DEF

UN PROBLEMA DI CAUCHY è BEN POSTO se

1. $\exists\, u$ sol
2. u è UNICA
3. u dipende in maniera CONTINUA DAI DATI DEL PROBLEMA (GRF PDE, g, BC, IC)

OS. PROBLEMA MAL POSTO PER EQ. ELLITTICHE

$$\begin{cases} \partial_{tt} u + \partial_{xx} u = 0 \\ u(0,x) = 0 \\ u_t(0,x) = \dfrac{\sin(kx)}{k} \end{cases} \quad \text{I.C.} \quad S = \{t=0\}$$

$$u(t,0) = u(t,2\pi) \qquad \text{PERIODICITÀ}$$